

SCAL Report — Well T1-31

Introduction

This report presents SCAL results for Well T1-31 in the Al-Jurf Offshore Field.

Six core plugs were analysed for relative permeability, capillary pressure, and wettability.

- Unsteady-state relative permeability (USS method)
- Mercury injection capillary pressure (MICP)
- Amott spontaneous imbibition wettability index

Results & Interpretation

All samples confirm water-wet behaviour. Amott indices range from 0.54 to 0.72.

- K_{rw} @ S_{or} : 0.072–0.142 (strongly water-wet)
- K_{ro} @ S_{wi} : 0.748–0.923 (excellent oil mobility)
- Estimated waterflood efficiency: 62–75%

Table 1 — Routine Core Analysis Results

Sample ID	Depth (m)	Porosity (%)	Kabs (mD)	Swi (%)	Amott Index
T131-01	2847.3	19.2	245.6	24.8	0.68
T131-02	2851.7	22.1	412.3	21.5	0.72
T131-03	2856.4	16.8	128.9	29.3	0.61
T131-04	2860.9	20.5	318.7	23.1	0.70
T131-05	2865.2	14.3	76.4	33.7	0.54
T131-06	2869.8	21.7	387.1	22.0	0.71